

a browser. This is quite useful in reusing code and even updating software. Although it does not have much to do with iOS mobile app development, it certainly can be on our list, particularly for apps for different platforms.

Python: Python is a high-level language that can be used to develop Android and desktop applications from scratch. It is an object-oriented language, which is processed at runtime by the interpreter. It's easy to learn, easy to read and is quite easy to scale. It supports all types of GUI apps and runs on Windows, Mac, UNIX and Linux.

Best open source mobile app development frameworks

As discussed in the beginning, mobile app developers make use of different frameworks in order to come out with bug-free and error-free apps. Here are some great open source frameworks.

Framework 7: This framework has been the top choice for the development of different iOS apps. Since it offers Android support as well now, it's actually a good option if we want to start with iOS and then build an Android version of an app, with an iOS look and feel. Its main advantage is that it helps developers to build different iOS applications with just JavaScript, CSS and HTML, which most app developers already know.

Features

1. Provides support for building different cross-platform hybrid mobile applications
2. Finest and rich framework with the most features for iOS hybrid application development
3. Material design UI
4. Native scrolling
5. 1:1 page animation
6. A custom DOM library
7. XHR caching and preloading

jQuery Mobile: This open source mobile application development framework has a large and committed user base in spite of the fact that it lacks many of the advanced features of most packages. This is because of its simplicity and other features like 'write once and run anywhere'. It's actually a good choice for the simple applications that need to run on Windows and BlackBerry phones. The main focus of jQuery is to empower its developers to build mobile apps that run seamlessly and with a unique user experience across all mobiles and tablets. It does not actually focus on providing the native look and feel to apps, for individual platforms like iOS and Android.

Features

1. Lightweight
2. Equipped with different theme designs
3. Semantic mark-up
4. Progressive enhancement
5. Phone Gap/Cordova support

Kendo UI: Kendo UI is a framework used for building



Figure 4: Features of Framework 7 (Image source: googleimages.com)

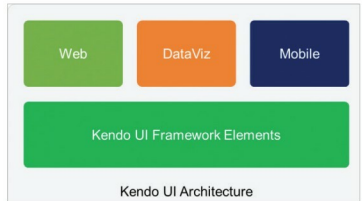


Figure 5: Kendo UI architecture (Image source: googleimages.com)

any kind of hybrid mobile application with JavaScript, HTML5 and CSS. It is available in open source as well as in commercial versions. It offers a wide range of UI widgets and plugins. It is being maintained and supported by Telerik, and has a large customer base, which includes different organisations like NASA, Sony, Microsoft, Toshiba and Volvo.

Features

1. Relies heavily on jQuery
2. Equipped with more than 70 ready-to-use jQuery widgets
3. Has numerous prebuilt themes added with Material Design styling, and a theme builder for customised options
4. Angular and Bootstrap UI integration
5. Performance can be optimised as per needs

Mobile Angular UI: This open source framework combines AngularJS and the modified version of Twitter's Bootstrap, and converts them into a mobile UI development framework. It supposedly retains most of Bootstrap 3's syntax for easier Web-to-mobile portability while adding different mobile components that are missing from Bootstrap, like overlays, switches, sidebars, fixed-positioned navigation bars and scrollable areas. It consists of different libraries like *overthrow.js* and *fastclick.js*. It is available for free use under the MIT licence.

Features

1. Provides the best of both the Angular framework and Bootstrap 3 to build different HTML5 mobile applications
2. No dependencies to jQuery or Bootstrap js
3. Uses *fastclick.js* and *overthrow.js* libraries for a smooth mobile experience